

MISSION 1 – WAYPOINT NAVIGATION

1.1 Objective

The objective of this mission was to configure and execute a multi-waypoint autonomous drone mission using NAV and DO commands while monitoring live telemetry data such as relative altitude and speed.

1.2 Mission Setup

Launch Point: Tech Green

Initial Height: 50 ft

Initial Speed: 30 mph

Waypoints:

1. Bobby Dodd Stadium – 52 ft
2. Couch Park – 54 ft
3. CRC – 56 ft
4. Van Leer – 58 ft
5. Coliseum – 60 ft
6. Russ Chandler – 62 ft
- Return to Launch – 50 ft
- Land

Height was incremented by 2 ft at every waypoint and restored before landing.

During execution in QGroundcontrol, the altitude can be seen in Meters. Because QGroundcontrol shows altitude in meters. We have converted the height in feet into meters. That means 50 ft = 15.2 m and incrementing of 2 ft for every next waypoint means that we have to add 0.61 m. The altitudes for waypoints becomes 15.2, 15.8, 16.5, 17.1, 17.7, 18.3 and so on.

1.3 DO vs NAV Commands

◆ NAV Commands

- Control movement of the drone
- Define location-based actions
- High priority — form the backbone of the mission
- Examples:
 - NAV_TAKEOFF
 - NAV_WAYPOINT
 - NAV_RETURN_TO_LAUNCH
 - NAV_LAND

NAV commands execute sequentially; the drone will not advance to the next step until the current NAV command is fully satisfied.

◆ DO Commands

- Perform auxiliary or behavioral actions during flight
- Execute within the context of the current NAV segment
- Lower priority — subordinate to NAV commands
- Examples:
 - DO_CHANGE_SPEED

Priority Statement: NAV commands define the mission's motion and path. DO commands modify the drone's behavior *during* NAV execution. The drone does not wait on a DO command to complete before continuing navigation — however, the exact behavior when a new NAV command is reached is **command-dependent**. Some DO commands (like DO_CHANGE_SPEED) take effect immediately and persist into subsequent segments, while others may be superseded when the next NAV waypoint is triggered.

Experimental Demonstration

The following mission command sequence was used to demonstrate the distinction:

WP1 – NAV_WAYPOINT → Bobby Dodd stadium @ 15.8 m, speed = 30 mph
DO_CHANGE_SPEED → speed set to 33.5 mph

WP2 – NAV_WAYPOINT → CRC
DO_CHANGE_SPEED

@ 16.5 m, speed = 33.5 mph
→ speed set to 35.7 mph

The drone did not stop or hover when DO_CHANGE_SPEED executed — it continued navigating while the speed change took effect. This confirms that **DO commands are non-blocking** and run alongside NAV execution rather than interrupting it.

1.4 Observations (Altitude & Speed)

- Relative altitude increased exactly by 0.61 m per waypoint as commanded.
- Speed stabilized around the commanded value between waypoints.
- During turns, speed briefly reduced due to heading correction.
- RTL correctly restored altitude to 15.2 m before initiating descent.

1.5 Conclusion

The drone successfully executed a structured multi-waypoint mission while maintaining accurate altitude increments and speed control. NAV commands governed the drone's path and motion sequencing, while DO commands modified in-flight behavior without interrupting navigation. The distinction between the two command types was confirmed experimentally: DO_CHANGE_SPEED altered groundspeed mid-mission without causing any pause in the flight path, demonstrating that DO commands are subordinate to, but concurrent with, NAV command execution.